

Matrices & Matrix Algebra



Matrix addition and scalar multiplication

- 1 Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 2 & 1 \end{bmatrix}$. Find $A + B$ and $2A - B$.
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Matrix multiplication

- 2 Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$. Find AB .
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Matrix multiplication is not commutative

- 3 For A, B above, compute BA and compare it to AB .
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The identity matrix and matrix inverses

- 4 Write the 2×2 and 3×3 identity matrices.
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Invertibility of 2x2 matrices

- 5 State the formula for the inverse of $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$.
- 6 Find the inverse of $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.
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Inverse of a product

- 7 State the formula for $(AB)^{-1}$ in terms of A^{-1} and B^{-1} .
- 8 If $A^{-1} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, find $(AB)^{-1}$.
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More matrix multiplication practice

- 9 Let $A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}$ (2×3) and $B = \begin{bmatrix} 3 & 1 \\ 2 & 0 \\ 1 & -1 \end{bmatrix}$ (3×2). Find AB .

Matrix-vector multiplication

- 10 Let $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$. Find $A\mathbf{v}$.
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The transpose

- 11 For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ (2×3), write its transpose A^T (3×2).
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More 2x2 inverse practice

- 12 Find the inverse of $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$.
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Solving a matrix equation

- 13 Use the inverse found above to solve $2x + 5y = 4$, $x + 3y = 1$ (written as $A\mathbf{x} = \mathbf{b}$).
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Symmetric matrices

- 14 A matrix is symmetric if $A^T = A$. Is $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$ symmetric?

- 15 Is $B = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$ symmetric?
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Powers of a matrix

- 16 Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Find A^2 .
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The trace of a matrix

- 17 The trace of a square matrix is the sum of its diagonal entries. Find $\text{tr}(A)$ for $A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$.
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Diagonal matrices

- 18** A diagonal matrix has zeros everywhere except the main diagonal. Write the 3×3 diagonal matrix with entries $2, -1, 5$ on the diagonal.
- 19** For the D above, find D^2 .